

Experiment – 06

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Subject: Advanced Database Management System

Question 1: Daily Sales Comparison (Medium)

A fruit distribution company records the number of apples and oranges sold each day. For every date, exactly one record exists for apples and one record exists for oranges. Generate a daily sales comparison report showing the difference between the number of apples sold and the number of oranges sold on the same day, calculated as $\text{Difference} = \text{Apples Sold} - \text{Oranges Sold}$. Display the sale date and the calculated difference. The result should be sorted by sale date in ascending order.

Sales Table:

sale_date	fruit	sold_num
2020-05-01	apples	10
2020-05-01	oranges	8
2020-05-02	apples	15
2020-05-02	oranges	15
2020-05-03	apples	20
2020-05-03	oranges	0
2020-05-04	apples	15
2020-05-04	oranges	16

Example Output:

sale_date	diff
2020-05-01	2
2020-05-02	0
2020-05-03	20
2020-05-04	-1

Solution:

```
SELECT
  A.SALE_DATE,
  (A.SOLD_NUM - O.SOLD_NUM) AS DIFF
FROM SALES AS A
JOIN SALES AS O
  ON A.SALE_DATE = O.SALE_DATE
  AND A.FRUIT = 'apples'
  AND O.FRUIT = 'oranges'
ORDER BY A.SALE_DATE ASC;
```

Question 2: Tree Node Classification (Hard)

A company stores hierarchical employee reporting information in a table. Each record represents a node in a hierarchy where id represents the current node and p_id represents the parent node. Classify every node into one of the following categories: Root — a node that does not have any parent; Leaf — a node that has a parent but does not have any children; Inner — a node that has both a parent and at least one child. Display the node ID along with its category, sorted by node ID.

Tree Table:

id	p_id
1	NULL
2	1
3	1
4	2
5	2

Example Output:

id	type
1	Root
2	Inner
3	Leaf
4	Leaf
5	Leaf

Solution:

```
SELECT
  ID,
  CASE
    WHEN P_ID IS NULL THEN 'Root'
    WHEN ID IN (SELECT DISTINCT P_ID FROM TREE WHERE P_ID IS NOT NULL)
      THEN 'Inner'
    ELSE 'Leaf'
  END AS TYPE
FROM TREE
ORDER BY ID;
```